

The 5-Layer Claude Code Memory Architecture

CLAUDE.md is not a memory system. Here is the full stack.
github.com/keshavsuki/recall-stack

@keshavsuki

Why most setups fail

Most people set up CLAUDE.md, call it done, and wonder why Claude keeps forgetting things.

CLAUDE.md is a rules file. It tells Claude who it is. It does not remember what you did yesterday, what went wrong last week, or what your knowledge base contains.

Most people have layer 1. Some have layer 2. Almost nobody has all five.

Recall Stack: all 5 layers in one repo

github.com/keshavsuki/recall-stack

One clone. One setup script. All five layers wired in automatically through hooks. Everything works every session without you doing anything.

Layer 1 CLAUDE.md

`~/.claude/CLAUDE.md`

Permanent rules and preferences. Loads every session automatically. Tells Claude how to behave, what tools to use, what to avoid.

Does NOT: Remember anything. Update itself. Learn from mistakes.

This is your foundation. Not your memory system.

Layer 2 primer.md

`~/.claude/primer.md`

Rewrites itself after every completed task. Active project, what was completed, exact next step, open blockers. If you close the terminal mid-session, the last completed task state is already saved.

Layer 3 Git Context

`SessionStart` hook

Injects live codebase context at every launch. Current branch, last 5 commits, modified files. All pulled fresh via the `SessionStart` hook. Claude knows what changed before you say a word.

Layer 4 Hindsight

`SessionStart` + `SessionEnd` hooks

Extracts behavioral patterns from past sessions and changes how Claude responds in future ones. `SessionEnd` hook retains patterns. `SessionStart` hook recalls them. Not retrieval. Actual learning.

Layer 5 Obsidian Vault

Shell alias with `--add-dir`

Your entire Obsidian vault becomes Claude's working directory via `--add-dir`. Every note you have ever written is now context. No plugin needed. Syncs to mobile via iCloud or Obsidian Sync.

Install: One Command

This installs layers 1, 2, 3, and 5 automatically. Layer 4 (Hindsight) is guided separately in the next step.

```
git clone https://github.com/keshavsuki/recall-stack.git
cd recall-stack
bash setup.sh --obsidian ~/path/to/your/vault
```

What `setup.sh` does

- Copies `CLAUDE.md` and `primer.md` to `~/.claude/`
- Copies hooks to `~/.claude/hooks/` and makes them executable
- Writes `SessionStart` and `SessionEnd` config to `~/.claude/settings.json`
- Adds the `--add-dir` alias for your Obsidian vault to your shell
- Optionally copies the post-commit hook to your current repo

Layer 4: Hindsight Setup

Why Docker for Hindsight?

Hindsight requires `uvloop`, which only runs on Linux. Docker runs a Linux VM under the hood so it works on Mac, Windows, and Linux. `pip install hindsight-all` will NOT work on Windows. Mac may work (`uvloop` is Unix-compatible) but Docker is the safest option on all platforms.

Step 1: Install Docker Desktop from docker.com/products/docker-desktop

Step 2: Run Hindsight as a persistent background container:

```
export ANTHROPIC_API_KEY=your-key-here
```

```
docker run -d --name hindsight --restart unless-stopped -p 8888:8888 -p 9999:9999 -e  
HINDSIGHT_API_LLM_PROVIDER=anthropic -e HINDSIGHT_API_LLM_API_KEY=$ANTHROPIC_API_KEY  
-v hindsight-data:/home/hindsight/.pg0 ghcr.io/vectorize-io/hindsight:latest
```

Step 3: Create the memory bank (one-time only):

```
curl -s -X PUT http://localhost:8888/v1/default/banks/claude-sessions \  
-H 'Content-Type: application/json' \  
-d '{"name": "claude-sessions"}'
```

That is it

The hooks in recall-stack handle retain and recall automatically.
SessionEnd retains patterns. SessionStart recalls them.
No extra wiring. No Python scripts. It just works.

The --restart unless-stopped flag means Hindsight starts automatically
when Docker Desktop opens. No manual restart needed after a reboot.
To stop it: `docker stop hindsight`
To start it again: `docker start hindsight`

Hindsight UI: `http://localhost:9999`
Hindsight API: `http://localhost:8888`

Supports: anthropic, openai, gemini, groq, ollama, lmstudio

Without Hindsight

Don't want to run Docker? The hooks skip Layer 4 gracefully if Hindsight isn't running. You still get layers 1, 2, 3, and 5. That is already better than what most setups have.

How It Works at Launch

You type: `claude`

Layer 1	CLAUDE.md loads	-->	how should I behave?
Layer 2	primer.md loads	-->	where did we leave off?
Layer 3	SessionStart hook fires	-->	what changed in the codebase?
Layer 4	Hindsight recall fires	-->	what patterns should I follow?
Layer 5	Obsidian vault mounts	-->	what is in the knowledge base?

Claude sees everything before you type a word.

...you work...

```
...primer.md auto-rewrites after each completed task...
...post-commit hook logs every commit...

Session ends:
Layer 4  Hindsight retain fires  -->  extracts behavioral patterns
                                         stored for next session
```

Summary

Layer	What it does
CLAUDE.md	Permanent rules. Never changes. You edit manually.
primer.md	Current state. Overwrites after every completed task.
Git Context	Branch, commits, modified files. Fresh every launch.
Hindsight	Behavioral learning. Learns automatically over time.
Obsidian	Full knowledge base. All your notes as context.

The repo

github.com/keshavsuki/recall-stack

Built by @keshavsuki. Powered by Hindsight for behavioral learning.
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